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Hiring Manager
Fireloop AI

Dear Hiring Manager,

I am writing to apply for the Forward Deployed Engineer (Robotics) position at Fireloop AI. As a UC Berkeley M.Eng. candidate in robotics graduating this May, I bring ROS2 engineering, embedded system integration, and field-level debugging experience—the core of what Fireloop's FDE role demands to move research-grade robotics into reliable client deployments.

My closest match to Fireloop's core stack is the L.I.F.T. project, where I built a ROS2-based robotic system coupling a simulator with an MPC controller for obstacle avoidance and trajectory completion. To close the control loop, I integrated an OptiTrack motion capture system delivering sub-millimeter position accuracy at under 10 ms latency for sub-centimeter trajectory tracking during iterative controller tuning. That full pipeline—simulation, controller, real-time sensor feedback—parallels the deployment workflow Fireloop executes when moving robotics stacks from simulation into client environments. I have not yet worked with Isaac SIM or Isaac Lab, but this simulation-to-hardware integration experience provides a direct ramp into Fireloop's toolchain.

Fireloop's FDE is the technical voice of the field, and that demands someone who debugs systematically and delivers under pressure. At Xiaoxian Intelligent, a third-party integrator's Siemens PLC code was failing intermittently on an industrial inspection line. I traced the fault through ladder logic and I/O timing, validated each automation stage end-to-end, and documented the root causes for the core engineering team—exactly the diagnose-and-escalate cycle Fireloop's FDE performs between client sites and headquarters. During the ROBOCON competition, I modularized an omnidirectional robot into decoupled gimbal, power, launcher, and chassis subsystems with redundant interfaces, specifically to isolate faults and accelerate system debugging—then delivered two autonomous platforms and one quadruped under strict fabrication deadlines.

My cloud and edge deployment experience is limited, but my embedded communication work—ESP32 Wi-Fi multi-device coordination, Modbus RTU gateways, CAN and UART protocols—provides a networking foundation to grow into containerized deployment quickly. I would welcome the chance to discuss how my integration and debugging background can serve Fireloop's client delivery mission.

Sincerely,
Felix Hu